

KAVYA KALAVALA

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PROFESSIONAL SUMMARY

Computer Science (Data Science) undergraduate with practical experience in Python-based data pipelines, AI/ML workflows, RAG evaluation, REST API concepts, JSON/CSV processing, database-backed applications, model evaluation, and technical documentation. Comfortable translating open-ended requirements into structured, testable prototypes involving data ingestion, transformation, validation, retrieval, summarization, quality checks, and handover-ready artifacts. Interested in data engineering, AI-enabled workflows, semantic search, ML evaluation, and reliable product-oriented systems.

TECHNICAL SKILLS

Programming and CS: Python, Java, C++, C, OOP, data structures, algorithms, debugging, documentation, Git/GitHub, Linux, Jupyter, VS Code.

Data engineering and software: ETL concepts, data ingestion, transformation, validation, metadata handling, data quality checks, SQL, DBMS, REST APIs, JSON/CSV payloads, workflow automation, logging and monitoring concepts.

AI/ML and NLP: machine learning, deep learning, supervised/unsupervised learning, feature engineering, model evaluation, retrieval-augmented generation, semantic search, summarization, classification, prompt engineering, and evaluation datasets.

Libraries and tools: NumPy, pandas, SciPy, scikit-learn, XGBoost, PyTorch, TensorFlow/Keras, Matplotlib, Seaborn, NetworkX, OpenCV, FastAPI, Django REST Framework, MySQL, LaTeX, Excel, Power BI/Tableau basics.

Professional skills: analytical thinking, problem solving, written communication, collaboration, literature review, presentation, independent ownership, and fast learning.

SELECTED PROJECTS

Enterprise Knowledge Copilot | *FastAPI, React, TypeScript, RAG, REST APIs, JSON, Azure-style search concepts*: Built a document-ingestion and Q&A prototype with upload, parsing, chunking, retrieval, citations, document-scoped querying, prompt-preview modes, query history, Hit@K, MRR, groundedness, latency, and review flags.

Context-Aware RAG Assistant and Evaluation Toolkit | *Python, retrieval, chunking, citations, evaluation*: Implemented a local RAG workflow for sample documents using chunking, TF-IDF/vector-style retrieval, grounded answer generation, source citations, Hit@K, MRR, answer recall, and groundedness scoring.

Smart Resource Allocation System | *Django, Django REST Framework, MySQL, APIs, dashboards*: Developed a booking system with catalog management, overlap prevention, role-aware workflows, REST serializers, JSON responses, utilization summaries, pending requests, and dashboard analytics.

AI GTM Lead Enrichment Automation Prototype | *Python, CSV, JSON, prompt payloads, webhook-ready automation*: Processed CSV lead records, assigned segments and rule-based priority scores, exported CRM-ready JSON, and generated LLM prompt payloads and Zapier-style webhook payloads.

PIRGE CodeGuard | *Python, reasoning graphs, policy checks, audit artifacts*: Built a pre-execution coding-agent review prototype using provenance graphs, action warrants, capability scoping, command-intent checks, allow/block decisions, audit JSON, and decision graphs.

Model Calibration and Overconfidence Diagnostics | *Python, scikit-learn, reliability diagrams, uncertainty*: Evaluated confidence behavior using ECE, Brier score, reliability diagrams, class-wise calibration, noisy/OOD-style settings, and temperature scaling.

MOGA-FCM Cognitive Load Detection | *Python, NSGA-III, Bayesian optimization, Fuzzy C-Means*: Combined multi-objective feature selection with Bayesian-optimized Fuzzy C-Means for physiological clustering and compared silhouette, feature-count objectives, baselines, and driver-level splits.

EEG Topology and Brain-Network Dynamics | *Python, graphs, persistent homology, ML evaluation*: Built functional-connectivity and topology-feature workflows for workload-related EEG structure analysis using graph descriptors and persistent-homology concepts.

DepMap / LUAD Biomedical AI Analysis | *Python, biomedical data, graph-sparse attention, validation tables*: Explored module-level biomedical features, graph-sparse attention, stability checks, ablation outputs, and validation tables for relapse-related analysis.

Gluco-Adapt Healthcare AI | *Python simulation, uncertainty, dashboard outputs*: Simulated risk-triggered non-invasive glucose sensing with physiological features, uncertainty estimates, dashboard summaries, and power/accuracy trade-off views.

Preferential-Attachment Network Research | *Python, graph simulation, likelihood estimation, finite-size diagnostics*: Developed graph simulation workflows for degree-collision reconstruction and inverse-parameter inference with plots, tables, and scaling diagnostics.

EDUCATION

Manipal Institute of Technology, Bengaluru | B.Tech in Computer Science (Data Science) | 2023-2027 | CGPA: 9.08/10.00 up to 6th semester

Deeksha Centre for Learning, Bengaluru | Pre-University, Science | 2021-2023 | Score: 96%

Appollo National Public School | Secondary Education | 2017-2021 | Score: 96%

CERTIFICATIONS

IBM Enterprise Grade AI and Cloud Solution; Cisco Cybersecurity Essentials and Packet Tracer; Infosys Springboard Introduction to AI, Deep Learning and Computer Vision; Coursera Deep Learning Specialization, TensorFlow ANN, IBM Machine Learning, Python Developer, GUI Programming in Python, HTML/CSS/JavaScript; Udemy Feature Engineering in Machine Learning.